

RECENT VIEWS IN THE FOUNDATIONAL CONTROVERSY IN STATISTICS

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Summary: The controversy between the classical and Bayesian approaches to statistics is reviewed with the emphasis on recent developments. A summary of the main differences is given, followed by a discussion of the main points in the continuing argument. The various attitudes towards the controversy are stated and evaluated. The connection with Kuhn's theory of scientific revolutions is discussed.

1. INTRODUCTION

During the first part of the twentieth century, statistics emerged to become a full-fledged scientific discipline. However, its growth was accompanied by a continuing controversy concerning its foundations. In this controversy there are two major positions to be distinguished, but there are various shades of opinion between these as well. These two major positions are the so-called classical point of view and the rival Bayesian point of view. While probably the majority of statisticians have been trained in the classical spirit and are happily carrying on along the same line, a growing and vocal minority has

adopted the Bayesian standpoint and is pushing hard for its case to be accepted.

The resulting situation is a serious one for the statistical community. Although few would go so far as Kempthorne (1971, p. 470, 471) in the following quotation, the quotation does serve to illustrate the seriousness with which the situation is viewed by some:

We have in the profession of statistics as in all other aspects of society internal frictions of an order of magnitude quite unknown in the past. Some internal friction is essential. But we seem to be approaching rapidly the situation in which we are losing any communality of outlook and are unable to present any sort of unified picture of our subject. Groups of individuals in statistics and in professional societies appear to be quite incapable of communicating with each other. A phenomenon was labelled very aptly by Dantzig in the early 50's as statistical priesthood... We seem to be rapidly approaching the case in which everyone is a priest with his own private religion... I believe there is no area with comparable disunity, with the exception of philosophy. The analogy is not surprising, perhaps, because the boundary between statistics and philosophy, particularly the philosophy of knowledge, seems very vague.

The following remark by Barnett (1982, p. 22) also illustrates the situation:

But whilst the personalistic element in the study of statistics acts as a stimulus for its development, it also has inherent danger. Human nature being what it is, there is always the risk that bigoted and extreme attitudes develop, and that argument descends to the emotional level of an interdenominational dogfight... Fortunately, little expression of this is apparent, but we might remark on a tendency there has been for factional groupings to develop...

In recent years the stage has been reached where the major differences could be pinned down and discussed at symposia, e.g. Godambe and Sprott (1971) and Harper and Hooker (1976). These were also discussed in books by Barnett (1982) and

Kyburg (1974) as well as in a large number of articles in journals, some of which will be referred to in this paper.

The purpose of this paper is to try and give a review of the main points of discussion in order to obtain a perspective on the recent trends in the area. While no claim to completeness will be made, a serious attempt is made to catch the spirit of the arguments, to point out the diversity of standpoints and to evaluate the various attitudes towards the controversy.

2. REVOLUTIONS IN STATISTICS?

The beginning of Bayesian statistics can be traced back to the publication of Bayes' Theorem in 1763 (cf. Seal (1978) for a short history and content of the paper). About 100 years later Laplace published his own version of Bayes' Theorem without referring to Bayes. He then freely used the result for inferential purposes and his stature as mathematician removed for several decades any doubt about the possibility of such usage. Especially on the European continent, the methods of Laplace received support, but there was strong criticism in England from people like Boole, Venn and Fisher (Jaynes (1979) gives a broad discussion). Perhaps this difference of opinion can be explained if one takes into consideration the philosophical empiricist tradition of the time in England and the more rationalist continental scientific philosophy. Boldrini (1972, p.137) remarks:

One can see here the different mental outlook of the two: the Englishman who derives his way of thinking from the tradition of Locke and Hume, and the Frenchman who aligned with the development of Cartesian rationalism.

Criticism from such an influential figure as Fisher, accompanied by the rise of the Neyman-Pearson theory, thrust Bayesianism

into the background during the first half of this century. However, a small number of supporters remained "carrying the Bayesian torch" (Good, 1980, p. 23). These were people such as Jeffreys, L.J. Savage, de Finetti and Lindley. They gradually gathered a following which grew to a strong factor in the statistical community, stating their case strongly and with conviction.

As pointed out in the Introduction, the resurgence of Bayesian statistics has led to intensive discussion of foundations in modern statistics. This strong difference of opinion has recently been viewed in the framework of crisis in science, especially crisis in physics, as discussed by Kuhn (1962) in his very influential book *The Structure of Scientific Revolutions*. This was done, for example, by Lindley (1980) who sees L.J. Savage as a main figure and by Good (1980) who discusses the contribution of Jeffreys. The question arises: is it reasonable to use the Kuhn framework?

Kuhn views the historical development of science as follows. A science is established in history as a science when the community of scientists involved agree about the framework within which they will carry on their scientific work. This requires agreement about what will be acceptable as valid problems, valid methods, etc. Such a framework Kuhn names a *paradigm*. A scientific community then consists of those who share a paradigm. Once a paradigm is established, the scientists concerned do their scientific work within the boundaries of the paradigm and this is called *normal science* by Kuhn. To him normal science can be characterized as *puzzle-solving*, which is comprised of refinement and extension of the theories within the paradigm. After a usually prolonged period of normal science it may happen that anomalies appear which may not be reconciled with the existing paradigm. These anomalies gradually accumulate and result in a *crisis*. A typical example of such a crisis is the discovery of certain anomalies in Newtonian

physics at the beginning of this century and which led physicists to fundamentally re-examine the Newtonian paradigm within which they were operating.

In Kuhn's view the response of a scientific community to such a crisis is a very interesting phenomenon. At first there is a general reluctance to seriously consider the anomalous discoveries, but as the effect of the anomalies become more serious, attempts are made to accommodate and explain them within the existing paradigm. Gradually some begin to admit that the anomalies have to be explained in a different way and a new paradigm emerges, which can account for the anomalies. However, the acceptance of the new paradigm is a very difficult process. It requires an entirely new orientation and the acceptance can be likened to a *conversion* (Kuhn's terminology) since it is not based on logic or experiment, but on more intangible grounds, such as intuition. The new paradigm is then entirely different from the old one - they are *incommensurable*. Acceptance of the new paradigm comes because of agreement by at least the majority of the scientific community involved. However, some resistance usually remains, and eventually the new paradigm only triumphs because those who resist disappear from the scene. Because of these characteristics of a paradigm change, Kuhn names the process a *scientific revolution*. A typical example of such a revolution is the change from Newtonian physics to the physics of Einstein at the beginning of this century.

In view of this short summary of Kuhn's description of scientific change, it is now possible to see how Lindley (1980) applies this to statistics and to evaluate his attempt.

For Lindley, as one of the most outspoken advocates of Bayesian statistics, the established paradigm is classical statistics as developed by people like Fisher, Neyman, E.S. Pearson and Wald. He points out the existence of curious confidence intervals and tests, which are the anomalies which threaten the existing

paradigm. However, as Kuhn describes, also in statistics the proponents of the existing paradigm choose to ignore these disturbing examples and refuse to question their basic framework. The emerging paradigm is then the Bayesian point of view. In a somewhat optimistic manner, Lindley adds (p. 8) (about L.J. Savage):

It is unfortunate that he did not live to see it finally triumph and provide the paradigm for normal statistics as some of us think it will.

The attempt to explain the controversy in statistics in terms of the philosophy of Kuhn is probably an exaggeration, but it is interesting and has advantages. An important advantage is that the dispute between the main statistical schools can be viewed as a very human matter and it seems reasonable to expect that the solution of the dispute might not be obtained as a result of pure logic or experiment, but by different means, such as persuasion. The personal nature of the differences of opinion has also been stated by others, e.g. Barnett (1982, p. 19):

...many aspects of the study of statistical inference and decision-making must involve the personal judgement of those who instigate or develop the different approaches. This means that criteria may be correspondingly 'arbitrary' in depending on such personal attitudes for their justification, and also that criticism and controversy are inevitable.

At present the conflict is still very much alive and to appreciate its nature it is necessary to give a discussion of some of the main aspects of the dispute. This will be done in the next section. As will be seen, it is not quite clear that we are in the midst of a revolution and that the Bayesian viewpoint will be triumphant. Too many very convincing points of criticism against this approach remain. The main picture is one of valid criticism on both sides and a need for the clarification of the various issues.

Insight can also be gained by viewing the whole matter against the background of philosophy and history. See e.g. Kempthorne (1976a) and Pearson (1978). Such an investigation is not attempted in this paper, but the reader is referred to Geertsema (1983).

3. THE NATURE OF THE CONTROVERSY

We start this section by summarizing and contrasting some of the main characteristics of the major schools of thought. In doing this, the danger of over-simplification exists, but one has the advantage of focusing on some important points.

Classical statistics, then, takes the following position:

- (1) The probability concept used, is the frequency concept. Properties of procedures are thus always interpreted in terms of "the long run".
- (2) Bayes' theorem is not acceptable as an instrument for inference, except in cases where all probabilities involved have a frequency interpretation. Techniques of Bayesian statistics are considered useful as theoretical aids, but are not to be used in practical applications.
- (3) The data for use in the inferential process are sample data. Prior information is not considered part of the data to be analyzed, except, perhaps, in an informal way. The reason for this exclusion lies in the difficulty of the quantification of prior knowledge which is often vague. On the question of whether information on the possible consequences of inference should be included in the analysis, there is difference of opinion. Neyman originally excluded this type of information, but later on emphasized the connection between statistical inference and statistical decision theory, a key aspect of which is the quanti-

fication and use of information concerning the consequences in the form of loss and risk functions.

- (4) The main problems are those of point estimation, interval estimation and hypothesis testing. A central concept is that of a *parameter*, which is a fixed unknown constant number (or vector) (cf. Stigler (1976)), the object of estimation and testing.
- (5) The procedures are based on statistics, which can be viewed as random variables, since sampling is in principle indefinitely repeatable and the statistics can vary from sample to sample. The *sampling distributions* of these random variables are of basic importance, since they describe this variation and can be used in evaluating the performance of statistics.
- (6) There is a series of criteria for the evaluation of procedures. Where possible, optimal procedures are to be found. Otherwise, the class of procedures must be limited by a criterion such as unbiasedness or invariance and optimality within the class investigated. The various criteria do not give the impression of a unified system, which is a source of criticism.

Before contrasting the two major schools, it is appropriate to give a short discussion of the modern view of Bayesian statistics as expounded, for instance, by Lindley (1976). A distinction between Bayesian statistics and classical statistics, is the series of attempts by Ramsy (1926), Savage (1954), De Groot (1970) and others to axiomatize this theory. The axioms are simple and intuitive and are formulations of requirements for rational behaviour of a decision maker. These requirements usually are stated as requirements of *consistency* and *coherence* of behaviour. From the axioms follow, as theorems, the existence of a probability measure on the parameter space (which portrays the beliefs of the decision maker concerning the different

parameter values) and the existence of a utility function which gives the preferences of the decision maker between the various decisions in terms of the expected value of the utility with respect to the probability distribution mentioned earlier. The decision maker then has to choose the decision which maximizes the expected utility. This way of acting is then *inevitable*.

The connection between this decision theoretic approach and statistics is that, according to this point of view, statistical inference is always done with decision making in mind. The axiomatic treatment then implies that statistical inference always should proceed by way of the posterior distribution on the parameter space, given the data. The posterior distribution has to be computed via Bayes' theorem, and this requires knowledge of the prior distribution, which is seldom available. Lindley (1976, p. 35)) states, however, that the difficulty about the prior distribution is exaggerated and that an effort should only be made to measure it. On the other hand, the advantages of the Bayesian approach are important to him, a major advantage being that, for the first time in statistics, it provides a formal system into which the subject fits. This makes the subject much easier to teach.

For purposes of contrasting with classical statistics, we now list some of the main points of Bayesian statistics in the same sequence as was done with classical statistics earlier in this section:

- (1) The probability concept used is a subjective one. Thus probability statements reflect the beliefs of the statistician in the unique, given situation.
- (2) Bayes' theorem is the key to all statistical inference, since this allows the computation of the posterior distribution. There are two main points of view concerning the determination of a prior distribution (cf. Efron, 1978,

p. 237), an objectivist and a subjectivist. In the objectivist approach, a so-called "flat" prior is used to portray a situation of ignorance or neutrality concerning the possible parameter values. In the subjectivist approach one tries to obtain a suitable prior by introspection and imaginary bets.

- (3) All available information is utilized in a Bayesian analysis. Prior knowledge is quantified and used via the prior distribution, observed data enter via the likelihood function and information concerning the consequences of decisions are used in the specification of the loss function. The process of quantification may be problematic, but should not be shied away from.
- (4) The main problems are the same, but the emphasis is on decision making as the all embracing problem. Also the problems are interpreted differently, e.g. in terms of subjective probabilities.
- (5) Posterior probabilities play a decisive role in the procedures, instead of statistics and sampling distributions.
- (6) There are no criteria for the evaluation of procedures. The analysis has a fixed pattern via the posterior distribution.

In view of the clear differences between the two schools it is not surprising that a major debate between the proponents of the two schools is in progress. The following is an attempt to group the arguments into a few main themes.

- (1) THE INEVITABILITY ARGUMENT. As was discussed above, it is the Bayesian point of view that in order to act coherently, the Bayesian approach is inevitable. This follows, on the one hand, from the axiomatic approach. On the other hand, this is reinforced by results from Wald's decision

theory, from which follows that the only admissible decision functions are those which maximize the expected utility. Thus use of an admissible decision function presupposes the acceptance of a prior distribution. (De Finetti, 1974, p. 119-120.) Proponents of the Bayesian approach are indeed very outspoken about the logical necessity of their system, as some of the references cited in this paper will show. See also Cornfield (1969).

But there are many objections to these claims. Kempthorne (1972, p. 176) insists that not only the axioms in the system should be examined to see whether they are appropriate, but also the consequences should be examined to determine if they are reasonable or plausible. One should not only rely on the axioms, because acceptance of the axioms implies acceptance of all their consequences. For this reason it is very difficult to develop axioms. To him the theory has little value, because it is extremely difficult to specify the consequences of the possible actions. Also, the construction of a utility function provides problems almost impossible to solve.

In the same connection, Finch (1971, p. 187) doubts that the Bayesian argument is a faithful copy of the way an experimenter revises his opinion. This underlines the point that the Bayesian system is a prescriptive system which prescribes the behaviour of a "rational" man, but it is not at all clear what the relation with the actual behaviour of people is.

A further point of criticism, closely related to those mentioned above, concerns the relationship of Bayesian inference to reality. Cox and Hinkley (1974, p. 389) argue that the primary objective of statistical inference usually is to end up near the "true" model, rather than to be internally consistent.

The classical approach of comparing long-run performance with the "true" values of the unknown parameters links the analysis to reality. They prefer this concept and the criteria based on it to mere internal consistency.

In the same vein, criticism is applied to the use of the word *coherence*, which, together with *consistency*, is one of the key concepts in the Bayesian axioms. Barnard (1971, p. 298) objects against the use of this word with its emotional overtones:

The use of the word *unbiased* for an estimate whose expectation equals the parameter has been criticised similarly. It sounds bad to admit to being biased, but surely it is even worse to be incoherent! Yet we all recognise now that bias in the technical sense, is almost always present in good estimators, and I do not expect it will be long before we come to accept incoherence in the technical sense as a common feature of rational discourse.

Barnard notes that he perceives signs of incoherence among those who adopt the Bayesian approach. An instance of this is the use of different prior distributions for the same parameter in slightly different situations, because of the mathematical convenience of having a conjugate prior.

On another level, Barnard (1972, p. 4) objects against the dominant role assigned to decision making in the Bayesian system. He argues that the development of science is not representable as a sequence of decisions. It involves imagination, insight and creativity. It is always tentative. But in the Bayesian model, the approach is the reverse of all this. The set of alternative actions must be known, the consequences of each choice must be known and, having made a decision, the decision maker is committed.

Thus we see that the inevitability argument is far from being generally accepted.

- (2) SUBJECTIVITY AND OBJECTIVITY. The Bayesian position of leading figures such as Savage, De Finetti and Lindley, is, by their own admission, a subjective one. The determining factor is their use of subjective probability throughout, for example I.R. Savage (1968) writes "my probability" whenever probability statements appear. This means that the main goal of statistical analysis is held to be, not an attempt to describe the real world, but to provide a way for rational behaviour. Thus, statistical pronouncements reflect the rationally constructed opinion of the statistician, taking, of course, into account all pertinent information.

A focus point of criticism here is on the choice of the prior distribution and the effect of this choice on the Bayesian analysis. Kempthorne (1976, p. 222) emphasizes that it is not possible, in a given situation, to talk about *the* Bayes solution, since there is a Bayes solution associated with each choice for a prior distribution and there is no compelling way to produce a particular prior distribution by purely logical analysis. If the prior reflects the beliefs of the individual, also the result will be that of the individual and it is not clear why someone else would find this acceptable. Elsewhere, Kempthorne (1972, p. 176) raises the difficulty that after writing down his prior distribution, an experimenter might find that the experimental results he then obtains are in conflict with his prior. The question then is what should be done with these two pieces of information.

Hamaker (1977) views the subjective character of the analysis as a threat to the statistical profession. He argues as follows: If a statistician should help his employer to reach a decision, the prior distribution should be that of the employer and not that of the statistician. But the employer has an interest in the decision to be reached and

this may lead to bias in his subjective probabilities. The statistician is called in to help make an independent and objective assessment of the situation and his report can be used as objective evidence, but it will be under suspicion if a subjective and possibly biased prior is used.

Cox and Hinkley (1974, p. 387) are willing to admit that, in some situations, such as in some business decision problems, there may be very little real data available and that then assessment of information from other sources is important. The Bayesian approach may then be useful for clarifying an individual's opinion. If people can agree about a prior distribution, and if they understand its rational basis, it could be used in an analysis. However, they are opposed to the uncritical acceptance of a prior distribution for use in a rational discussion. A further objection they have is that prior information is often no more than an unspecified feeling and that it should not be treated in the same way as data which were carefully obtained through observation or experiment.

The Bayesian response to this criticism goes in two directions. First, it is argued that objectivity is present in the Bayesian approach. For instance, De Finetti (1974, p. 121) states

Such ideas are however distressing for some people, who consider objectivity, in the strictest sense, as a necessary attribute of probability and of science. But the regret for losing the faith in the perfect objectivity of probability, and hence of science, is unjustified. Nothing is lost but what was a mere illusion. And such illusory objectivity is now replaced by the effectual objectivity, that is the true degree of objectivity attainable by human science through human senses and mind.

He also mentions that different prior distributions lead to the same conclusion in the presence of a sufficiently large body of data. This is sometimes called the "principle of precise measurement" which implies that enough data "swamps" the prior information so that different individuals arrive at about the same conclusions (cf. Barnett (1982), p. 210, 228). Also, Jaynes (1976, p. 236) points out that if the principle of maximum entropy is included, it is possible to derive the correct prior distribution by logical analysis.

The second response from the Bayesian side is to contend that classical statistics is not so objective as is claimed by its supporters. Good (1976) claims that classical statistics has more success in hiding the subjective aspects. Jaynes (1976, p. 237) points out that in classical statistics various criteria operate, which are arbitrary and which lead different statisticians to different solutions. Lindley (1971, p. 303) states that, in classical statistics, the reference set is arbitrary:

I have recently become convinced that the most important difference between the two schools is that whereas a Bayesian uses as a prop for his methods, the prior distribution, the prop for the other approaches is the reference set. Bayesians are often attacked for the arbitrariness of their prior. A reply is that the choice of reference set involves similar arbitrary selections.

This latter criticism requires some explanation. In the classical theory the frequency view of probability requires that an experiment be viewed as one which could (at least in the imagination) be repeated indefinitely. This implies that not only the result obtained is important, but also the entire collection of possible results which could be obtained. This collection, called the sample space or reference set, is then the set over which the probability has to be distributed. The problem which arises is that this reference set is not unique and that, in principle,

an infinite number of reference sets are allowable in any given situation. Often there is a set which is intuitively the "correct" one, but there are realistic examples of situations where the choice of the reference set is not clear and can lead to confusion.

An example of this is originally due to Cox (1958), but has since then been discussed also by Cornfield (1969, p. 620), Kyburg (1974, p. 52) and Giere (1976, p. 68). In the example there are two sources of observations. One source is chosen at random and observations taken from it. Should the sample space provide for observations from both sources or just from the chosen source? Optimal procedures based on the first possibility may be intuitively unacceptable. In order to prevent this kind of difficulty, there have been attempts to establish a conditionality principle. But this has not been entirely successful (cf. Efron, 1978, p. 238-241).

An example which illustrates difficulties which can arise in choice of reference set, is due to Pratt (1962, p. 314-316) and has also been discussed by Efron (1978, p. 236-237). An observer makes a number of measurements and constructs a confidence interval. Afterwards he is told that the measuring instrument was defective and that measurements smaller than a certain value could not be made. Fortunately the measurements obtained are safe, because they are larger than the boundary value. However, the effect of the new information is that the sample space presumably should change to allow fewer possibilities and according to the criteria of classical statistics the confidence interval should then change also. What is intuitively unacceptable here is that something that *could* happen, but did not, exerted influence on the inference procedure. Bayesian statistics is free of this kind of contradiction, since it utilizes only what actually has been observed and not what could have

been observed.

Thus, both approaches are criticized for a lack of objectivity. But there is a difference. While there is general agreement that absolute objectivity is impossible, opponents of the Bayesian approach would limit this to the application of the mind and imagination of the statistician to the formulation of a model and a problem to be solved. The analysis to follow should be objective (cf. Kempthorne (1969, p. 654) and Cox and Hinkley (1974, p. 388)). They would agree with Efron (1978, p. 245):

But consistency isn't enough. Subjective Bayesianism must face the challenge of scientific objectivity. This is the ultimate stronghold of the frequentist viewpoint.

- (3) ACCUSATIONS OF DOGMATISM. So far we have discussed criticisms which amount to being incoherent, inconsistent and subjective - this is almost synonymous to a criticism of being unscientific. To this we now add a criticism in the same vein, namely that of being dogmatic - also an attitude commonly considered unscientific. The *Random House Dictionary* describes dogmatism as:

Unfounded positiveness in matters of opinion.
Arrogant assertion of opinion as truths.

Thus Neyman (1957, p. 19) objects against the forced use of an unconvincing prior distribution:

What I am opposed to is the *dogmatism* which is occasionally apparent in the application of Bayes' formula when the probabilities *a priori* are not implied by the problem treated and an author attempts to impose on the consumer of statistical methods the particular *a priori* probabilities invented by himself for this particular purpose.

Kruskal (1978, p. 891) comments on the attitude of a leading Bayesian, L.J. Savage:

Personal probability was not only useful and interesting to study; it became for him the only sensible approach to probability and statistics. Thus, orthodoxy of neoradicalism developed: if one were not in substantial agreement with him, one was inimical, or stupid, or at least inattentive to an important scientific development.

He characterizes his own attitude as *eclectic*, using of various approaches what seems suitable in a given situation. This seems a popular attitude nowadays.

Statements by other Bayesians also describe their approach as the only valid one, as the following quotations show:

...it is the only satisfactory approach to statistics amongst those currently available (Lindley, 1976a, p. 353). I have mentioned that the concept of coherence is absent from orthodox statistics. This absence means that Bayesian statistics is quite different from standard teaching...To do good Bayesian statistics it is best to forget the orthodox ideas (Lindley, 1976a, p. 360). ...the major problem of inference has been solved in its essentials by Bayesian ideas. Why is Bayesian inference correct? (Lindley, 1971, p. 435.)

One has the impression that such strong claims for one line of thought are not accepted well by the statistical community. For instance, Kempthorne (1976, p. 221-222) denies that there is any significant attitude of *orthodoxy* (dogmatism?) in working statisticians. Working statisticians use whatever methods come in handy from a variety of approaches.

- (4) QUESTIONS ABOUT INTERNAL CONTRADICTIONS. From both sides there is criticism concerning possible internal contradictions.

A source of controversy in Bayesian statistics is the popular use of improper prior distributions, which cannot really describe probabilities. The question that arises is how one can justify the combination of these improper priors with actual probability distributions through Bayes' theorem and get probabilities. Recently a stir was caused by Dawid, Stone and Zidek (1973) who point out an inconsistency which follows from the use of improper prior distributions. Defending improper priors, Jaynes (1980) attempts a resolution of this paradox, but this is denied by Dawid, Stone and Zidek (1980). Lindley (1976b) now admits a change of position and recommends that improper priors not be used.

A contradiction which was pointed out in the classical theory, is contained in an example of Lehmann (1959, p. 116, no. 29) of a most powerful test which is significant at a certain level, but not at a larger level. This casts doubt over the classical theory of testing in general (Pratt, 1961, p. 166). Cornfield (1969, p. 625) claims that a necessary and sufficient condition to avoid this kind of problem, is to follow a Bayesian argument.

Jaynes (1976) discusses examples of classical confidence intervals which give rise to contradictions and leads, in his viewpoint, to suspicion about the entire classical approach.

To conclude this section, it should be pointed out again that the impression should not exist that the statistical community is divided into two homogeneous groups, one supporting the classical position and the other the Bayesian position. Within each of these groups there is also difference of opinion and, furthermore, there are also other views, different from the two main views, such as the likelihood approach, the fiducial methods originated by Fisher and structural methods due to D.A.S. Fraser.

4. ATTITUDES TOWARDS THE CONFLICT

In the statistical community and amongst philosophers of science, there is a variety of opinions about the controversy in the foundations. The following is a summary of several of these opinions:

- (1) Both approaches make a valid contribution to statistics. Hopefully they will grow together, giving rise to a new synthesis (see Efron (1978), Good (1976), Cox (1980) and the ensuing discussion). The two approaches are complementary and different situations may call for different approaches (Dempster, 1971, p. 58). Eventually those techniques that work, will survive (cf. Kyburg, 1974, p. 148).
- (2) There should be a reconciliation and this should be done by reinterpreting methods of one approach in terms of the viewpoint of the other. If this could be done, statisticians will all be working in the same way, but there will be a difference in interpretation (Bartholomew, 1971, p. 418).
- (3) Conflict in the foundations is inevitable. The reason for this is to be found in the importance of personal attitudes and subjective value judgements. Criticism is often directed towards the arbitrariness of criteria or concepts, but what for one person is arbitrary, is for the other self-evident (Barnett, 1982, p. 21).
- (4) The controversy is healthy. It generates a constructively critical attitude towards the subject. The subject has grown tremendously in this climate and new directions are born because of the differences (Barnett, 1982, p. 21; Efron, 1978, p. 232).
- (5) It is a myth that there are foundations for science and hence the controversy about the foundations has no meaning. Although several sciences complain of foundational problems,

these sciences themselves are flourishing. The existing problems arise because of language difficulties, which have little effect on the subject itself. It would be better to concentrate on the subject itself and be descriptive rather than prescriptive (Bross, 1971, p. 510-512).

- (6) Critical study is needed. The problems of communication between the schools of thought is serious and statisticians should try to come closer to each other. This should happen through tolerance of different views and a willingness to subject these views to critical examination. There must be a readiness to admit error (Kempthorne, 1971, p. 470-471).

These attitudes are also subject to criticism. A basic question is whether the conflict is real or not. The survey in Section 3 contains strong evidence that it is real. Furthermore, examples exist which show that application of techniques of the different approaches to the same situation may lead to very different conclusions (e.g. Kyburg, 1974, chapter 6). What is more, while some conflict is inevitable and while the subject is able to grow and function, the present conflict is extensive and it is hard to imagine that statistics will be served by ignoring it. To argue, for instance, that methods that work well are those that will survive, is to argue in a circular way, because a judgement of which methods work well would have to be based on a method of statistical inference. Attempts to remove the conflict by mutual reinterpretation of techniques have also not been too successful.

Is statistics in the midst of a revolution? The vast majority of opinions discussed above seem to argue that we are not. Rather than looking forward to the victory of some approach, the main line of thinking seems to favour a coexistence and maybe a synthesis of the different

approaches. But we shall have to wait and see. In the meantime examination and clarification of the differences are important so that in statistical application and teaching there can be a realization of the nature of the activity. Statistical workers and students should know that there is no *correct* way of doing statistics, but that there are several approaches, each with its own underlying ideas and motivation and that these should be well understood to be able to use the available techniques.

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